

## Appendix

### Notation

#### Sets and Index

|              |                                                                  |
|--------------|------------------------------------------------------------------|
| $i \in I$    | Set of suppliers (domestic and international)                    |
| $j \in J_f$  | Set of manufacturing facilities for firm $f$                     |
| $k \in K_f$  | Set of DCs for firm $f$                                          |
| $k' \in K_g$ | Set of joint green-DCs (if activated)                            |
| $m \in M$    | Set of market regions (urban/rural Türkiye, Germany, Iraq, etc.) |
| $c \in C$    | Set of sales channels (e-commerce, retail, 3P sellers)           |
| $f \in F$    | Set of firms (Firm 1, Firm 2)                                    |
| $v \in V$    | Set of transportation modes                                      |
| $p \in P$    | Set of product types                                             |
| $r \in R$    | Strategy scenario (Non-coop, Cooperative, Hybrid Coopetition)    |
| $a \in A$    | Set of raw material assortments                                  |
| $t \in T$    | Time periods in planning horizon                                 |

#### Model Parameters

##### Demand, Pricing, and Service Parameters

|                                        |                                                                                                                                                          |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\theta_{mct}$                         | Base demand scale for market $m$ and channel $c$ in period $t$ , representing the maximum attainable demand in the absence of price and service effects. |
| $\beta_{mct}$                          | Own-price sensitivity coefficient for market $m$ and channel $c$ in period $t$ , capturing the negative impact of higher retail prices on demand.        |
| $\sigma_{fmct}$                        | Brand/service uplift factor for firm $f$ in market $m$ and channel $c$ in period $t$ , reflecting firm-specific competitive advantages.                  |
| $\eta_{mct}^{ff'}$                     | Cross-price sensitivity between firm $f$ and competitor $f'$ in market $m$ and channel $c$ in period $t$ , measuring the substitution effect.            |
| $O_{mct}$                              | Service penalty coefficient for market $m$ and channel $c$ in period $t$ , used in effective price calculations to capture service-level effects.        |
| $M_{fmct}$                             | Margin/price floor for nominal price                                                                                                                     |
| $\Delta p_{mc}^{max}$                  | Max inter-period price change for market mmm, channel $c$ . (Currency/period.)                                                                           |
| $\Delta_{fmc}^{ch}$                    | Max channel price gap (e.g., retail vs. e-commerce) for firm $f$ , market $m$ , period $t$ .                                                             |
| $\underline{p}_{fmct}, \bar{p}_{fmct}$ | Minimum and maximum allowable retail prices for firm $f$ in market $m$ and channel $c$ in period $t$ , imposed by policy or market constraints.          |

Service quality is captured using a proxy function with coefficients:

|                                |                                                                                                                   |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------|
| $\alpha_0, \alpha_1, \alpha_2$ | Coefficients for the service proxy function, capturing the impact of utilization and mode mix on service quality. |
| $\theta_v$                     | Mode-specific penalty coefficient in the service proxy calculation.                                               |

##### Supplier and Raw Material Parameters

|                                      |                                                                                                                          |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| $r_{ap}$                             | Material requirement coefficient, which specifies the quantity of material $a$ required per unit of product $p$ produced |
| $a_{iat}, b_{iat}$                   | Intercept and slope of wholesale demand for supplier $i$ , material $a$ in period $t$ .                                  |
| $\gamma_{ii' t}$                     | Cross-price slope between supplier $i$ and competing supplier $i'$ in period $t$ .                                       |
| $s_{iat}^{max}$                      | Maximum raw material supply capacity of supplier $i$ for material $a$ in period $t$ .                                    |
| $c_{iat}^{wh}$                       | Unit wholesale procurement cost from supplier $i$ for material $a$ in period $t$ .                                       |
| $\underline{w}_{iat}, \bar{w}_{iat}$ | Minimum and maximum allowable wholesale prices for supplier $i$ for material $a$ in period $t$ .                         |
| $c_{it}^{tariff}$                    | (fraction): import tariff for supplier $i$ (0 for domestic).                                                             |
| $f x_t$                              | (currency conversion): exchange rate (international suppliers).                                                          |
| $\lambda_q, \lambda_l, \lambda_e$    | Weights for quality/lead time/carbon in the award score.                                                                 |
| $\psi_l$                             | Lead-time penalty coefficient.                                                                                           |
| $q_{ai}$                             | Quality score of material $a$ from supplier $i$ .                                                                        |
| $L_{ait}$                            | Lead time from supplier $i$ for material $a$ in period $t$ .                                                             |

Policy and sourcing constraints:

|             |                                                               |
|-------------|---------------------------------------------------------------|
| $q_a^{min}$ | Minimum average quality of awarded suppliers for material $a$ |
|-------------|---------------------------------------------------------------|

|                                                           |                                                                                                                                               |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| $L_{at}^{max}$                                            | Maximum average lead time allowed for awarded suppliers of $a$ in period $t$ .                                                                |
| $\rho_{fa}^{dom} \in [0,1]$                               | Domestic sourcing share target for firm $f$ , material $a$ .                                                                                  |
| <b>Production, Inventory, and Distribution Parameters</b> |                                                                                                                                               |
| $c_{fjpt}^{prod}$                                         | Unit production cost for firm $f$ at plant $j$ producing product $p$ in period $t$ .                                                          |
| $\bar{Q}_{fjpt}^{prod}$                                   | Maximum production capacity for firm $f$ at plant $j$ for product $p$ in period $t$ .                                                         |
| $\Delta \bar{Q}_{fjpt}^{prod}$                            | Incremental production capacity expansion available for plant $j$ of firm $f$ producing product $p$ in period $t$ .                           |
| $\bar{Q}_{k't}^{DC,g}, \bar{Q}_{fkpt}^{DC}$               | Storage and handling capacity at DC $k$ and shared green-DC $k'$                                                                              |
| $c_{fkpt}^{Inv}, c_{fk'pt}^{Inv}$                         | Inventory holding cost at DC $k$ and shared green-DC $k'$ .                                                                                   |
| $\phi_{fk}, \phi_{fk'}$                                   | Inventory spoilage or shrinkage rate at DC $k$ and shared green-DC $k'$                                                                       |
| <b>Emissions and Carbon Policy Parameters</b>             |                                                                                                                                               |
| $e_{ia}^{sup}, e_{fjp}^{prod}, e_{vuh}^{Tran}$            | Emission factors for suppliers and production and transportation (arc $u \rightarrow h$ ).                                                    |
| $\rho_{ia}^{sup}, \rho_{fjp}^{prod}, \rho_{k'}^{Trans}$   | Emission reduction factor for suppliers and green production technology and shared green DC.                                                  |
| $\tau_t$                                                  | Carbon tax rate per tonne of CO <sub>2</sub> -equivalent emissions in period $t$ .                                                            |
| $\bar{E}_f$                                               | Total allowable cumulative emissions (cap) for firm $f$ over the planning horizon.                                                            |
| $\pi_t$                                                   | Carbon permit price in a cap-and-trade mechanism for period $t$ .                                                                             |
| $v_{ft}^{min}, v_{ft}^{max}$                              | Lower/upper bounds on net permit trades for firm $f$ in period $t$ (negative = sell).                                                         |
| $\hat{E}_{ft}$                                            | Allowed emissions (quota) for firm $f$ in period $t$ without buying permits                                                                   |
| <b>Investment and Financial Parameters</b>                |                                                                                                                                               |
| $F_{fjp}^{PL}, F_{fk}^{DC}, F_{k'}^g$                     | Fixed activation cost of plant $j$ , DC $k$ , and shared green-DC $k'$ .                                                                      |
| $K_{fjp}^{cap}$                                           | Investment cost for production capacity expansion at plant $j$ of firm $f$ for product $p$ .                                                  |
| $K_{fjp}^{green}$                                         | Investment cost for adopting green production technology at plant $j$ of firm $f$ for product $p$ .                                           |
| $\lambda_{fk'}$                                           | Cost-sharing weight for shared green-DC facilities $k'$ .                                                                                     |
| $B_{ft}$                                                  | Available budget of firm $f$ in period $t$ for investment.                                                                                    |
| $\delta_t$                                                | Inter-period discount factor for net present value calculations in period $t$ .                                                               |
| <b>Transportation Parameters</b>                          |                                                                                                                                               |
| $c_{vuht}^{Tran}$                                         | Unit transportation cost for mode $v$ along arc $u \rightarrow h$ in period $t$ .                                                             |
| $Cap_{vt}^{tran}$                                         | Capacity of transportation modes $v$ in period $t$ .                                                                                          |
| <b>Other Strategic Planning Parameters</b>                |                                                                                                                                               |
| $\chi_{fpmc}$                                             | The share or mapping of product $p$ of firm $f$ to market $m$ and channel $c$                                                                 |
| <b>Decision Variables</b>                                 |                                                                                                                                               |
| <b>Binary Variables</b>                                   |                                                                                                                                               |
| $y_{fjpt}^{PL}, y_{fkt}^{DC}, y_{k't}^{green}$            | Binary variable indicating whether plant, DC, shared green-DC is activated.                                                                   |
| $U_{fjp}^{cap}$                                           | Binary variable indicating whether production capacity at plant $j$ of firm $f$ for product $p$ is expanded.                                  |
| $z_{fjp}^{prod}$                                          | Binary variable indicating whether green production technology is adopted at plant $j$ of firm $f$ for product $p$ .                          |
| $y_{ifat}^{sup}$                                          | Binary variable indicating whether supplier $i$ contracts for firm $f$ for raw material $a$ in period $t$ is won.                             |
| <b>Continuous Variables</b>                               |                                                                                                                                               |
| $\omega_{iat}$                                            | Wholesale price set by supplier $i$ for raw material $a$ in period $t$ .                                                                      |
| $P_{fmct}$                                                | Retail price set by firm $f$ for market $m$ and channel $c$ in period $t$ .                                                                   |
| $q_{fjpt}^{prod}$                                         | Quantity of product $p$ produced at plant $j$ of firm $f$ in period $t$ .                                                                     |
| $x_{ifjat}^v$                                             | Quantity of raw material $a$ shipped from supplier $i$ to plant $j$ of firm $f$ using transport mode $v$ in period $t$ .                      |
| $x_{fjkpt}^v, x_{fjk'pt}^v$                               | Quantity of product $p$ shipped from plant $j$ of firm $f$ to DC $k$ shared green DC $k'$ and using mode $v$ in period $t$ .                  |
| $x_{fkmcpt}^v, x_{fk'mcpt}^v$                             | Quantity of product $p$ shipped from DC $k$ and shared green-DC $k'$ of firm $f$ to market $m$ and channel $c$ using mode $v$ in period $t$ . |

|                       |                                                                                                                                                                        |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $I_{fkpt}, I_{fk'pt}$ | Inventory level at DC $k$ and shared green-DC $k'$ of firm $f$ for product $p$ at the end of period $t$ .                                                              |
| $E_{ft}$              | Total CO <sub>2</sub> -equivalent emissions generated by firm $f$ in period $t$ .                                                                                      |
| $v_{ft}$              | Quantity of carbon permits purchased or held by firm $f$ in period $t$ .                                                                                               |
| $\Psi_{ft}$           | Define as the bank (inventory) of permits at the end of period $t$ for firm $f$ . That means End-horizon bank cap (or set $\Psi_{ft} = 0$ if permits must be cleared). |
| $D_{fmct}$            | Demand served by firm $f$ in market $m$ and channel $c$ in period $t$ , as determined by the demand function.                                                          |
| $P_{fmct}^{price}$    | Effective price for firm $f$ in market $m$ and channel $c$ in period $t$ , adjusted for service-level penalties.                                                       |
| $\ell_{fmct}$         | Service proxy value for firm $f$ in market $m$ and channel $c$ in period $t$ , derived from utilization and transport mode mix.                                        |
| $Q_{fiat}^{raw}$      | quantity firm $f$ buys from supplier $i$ for material $a$ in period $t$                                                                                                |
| $S_{fiat}^{sup}$      | award score used to pick the winning supplier.                                                                                                                         |

**Sketch of proof (3.2.1).** By strong duality,  $C_f^{oper}(P) = \max_{\lambda_f \geq 0, \varsigma_f} \min_{x_f \geq 0} \mathcal{L}_f(x_f, \lambda_f, \varsigma_f; P)$ . Danskin's theorem (Danskin., 1966; Al-Dujaili et al., 2019) for pointwise maxima of affine functions in the parameter gives  $\partial C_f^{oper} / \partial P = -(\varsigma_f^*)^\top \partial b_f(P) / \partial P$  for any dual optimizer  $\varsigma_f^*$ . Only the market-clearing rows of  $b_f(P)$  depend on  $P$ , and their duals are precisely  $\mu_{fmct}$ . The last equality follows by the chain rule because  $\partial b_f / \partial P_{fmct}$  is zero everywhere except at the  $(f, m, c, t)$ , entry, where it equals  $\partial D_{fmct} / \partial P_{fmct}$ . Economically,  $\mu_{fmct}$  is the all-inclusive marginal system cost of serving one extra demand unit in  $(m, c, t)$ , already internalizing production, transport, inventory, award, capacity, carbon and green-technology prices through the LL KKT system (Allende and Still., 2013; Kuhn and Tucker., 2013; Kjeldsen., 2000; Karush., 1939).

**Sketch of proof (3.2.1).** Differentiate  $\Pi_f$  in (54) with respect to  $P_{fmct}$ . The direct derivative of the revenue term is  $D_{fmct} + P_{fmct}^{price} \partial D_{fmct} / \partial P_{fmct}$ . The derivative of the LL value  $-\mu_{fmct} D_{fmct}$  equals  $-\mu_{fmct} \partial D_{fmct} / \partial P_{fmct}$  by Lemma 1. Adding both gives (56). Substituting the constant-elasticity derivative yields (57).

**Sketch of proof (3.2.1).** On any region of  $\wp$  where the LL active set is fixed,  $\mu$  is constant and the right-hand side of (57) is continuous and single-valued in  $P_{-f}$ ; projecting onto  $\wp$  preserves nonemptiness, convexity, and upper hemicontinuity. Finite unions of such regions preserve upper hemicontinuity (Berge's maximum theorem arguments (Berge., 1984)). Continuity comes from continuity of  $D_{fmct}$  in  $P$  and piecewise constancy of  $\mu_{fmct}$ . Strict concavity of  $\Pi_f$  in  $P_f$  on a fixed active set follows from the constant-elasticity form, since  $(P - \mu)P^{-\beta}$  has a unique maximizer when  $\beta > 1$ . Kakutani's fixed-point theorem (Glicksberg., 1952; Kakutani., 1941) applies on the compact convex  $\wp$ , yielding a fixed point.

**Sketch of proof (3.2.2).** In log-price space,  $\partial \Phi_{fmct} / \partial S_{fmct} = D_{fmct}(1 - \beta_{mct})$  and cross-partial  $\partial \Phi_{fmct} / \partial S_{f'mct} = D_{f'mct} \eta_{mct}^{ff'}$  come from (25). Diagonal dominance (58) makes the pseudogradient strictly monotone. The dependence of  $\mu_{fmct}$  on rivals' prices is weak (piecewise constant or nonincreasing under (A2)), preserving monotonicity. holds and the dependence of  $\mu_{fmct}$  on rivals' prices is insensitive (nonincreasing) (true on any fixed active-set region of the lower level), which holds piecewise under a fixed LL active set. Then the symmetric part of the Jacobian is negative definite, the pseudogradient is strictly monotone, and the price equilibrium is unique. It means Rosen's condition applies and yields uniqueness. Comparative statics follow directly from (57) discussed in Corollary 1.

**Sketch of proof (3.2.2).** Write  $\Pi^{Coal}(P) = \sum_{fmct} (P_{fmct}^{price} - \mu_{mct}^c) D_{fmct}$  up to price-independent terms. Differentiate w.r.t.  $P_{fmct}$  the direct term contributes  $D_{fmct} + P_{fmct}^{price} \partial D_{fmct} / \partial P_{fmct}$ , the envelope term subtracts  $\mu_{fmct} \partial D_{fmct} / \partial P_{fmct}$ , Cross-effects add  $\sum_{f \neq f'} (P_{f'mct}^{price} - \mu_{mct}^c) \partial D_{f'mct} / \partial P_{fmct}$ .

Substituting  $\partial D_{fmct}/\partial P_{fmct} = -\beta_{mct} D_{fmct}/P_{fmct}$  and  $\partial D_{f'mct}/\partial P_{fmct} = +\eta_{mct}^{ff'} D_{f'mct}/P_{fmct}$  and rearranging gives the stated markup.

**Sketch of proof (3.2.2).** In log-price space the coalition's revenue in each  $(m, c, t)$  is a strictly concave function when own-elasticity dominates the internalized cross-effects; the lower-level cost term does not depend directly on  $P$  beyond the envelope structure already differentiated out. Summing over firms preserves strict concavity, so the optimizer is unique. With the coordinated markup in Proposition 3, and as long as price bands do not bind, the comparative statics are

$$\frac{\partial P_{fmct}^*}{\partial \mu_{mct}^c} = \left[ \beta_{mct} - \sum_{f \neq f'} \eta_{mct}^{ff'} \frac{D_{f'mct} P_{fmct}}{D_{fmct} P_{f'mct}} \right]^{-1} > 0, \quad \frac{\partial P_{fmct}^*}{\partial \ell_{fmct}} = \frac{O_{mct}}{\beta_{mct} - \sum_{f \neq f'} \eta_{mct}^{ff'} \frac{D_{f'mct} P_{fmct}}{D_{fmct} P_{f'mct}}} > 0 \quad (70-A)$$

Therefore, sweeping or green activation that reduces  $\mu^c$  pushes down coordinated posted prices for a given markup denominator, while better service reduces  $P^{price} - P$  and hence allows lower posted  $P$ .

**Sketch of proof (3.2.2).** This is the standard capacity-investment complementary-slackness rule. The Lagrangian contribution of (67) is  $\psi_{k't}^g \left( \sum_{fmc} x_{fk'mc}^v - \bar{Q}_{k't}^{DC,g} y_{k't}^{green} \right)$ . At optimum, the marginal value of one more unit of capacity is  $\psi_{k't}^g$ . The present value of the capacity benefit is  $\sum_t \delta_t \bar{Q}_{k't}^{DC,g} \psi_{k't}^g$ . If this exceeds the present value of the fixed activation costs, the KKT condition on  $y_{k't}^{green}$  prefers switching it on; if the shadow value is zero everywhere, there is no gain from activation.

**Sketch of proof (3.2.3).** For constant elasticity demand  $D(P) = kP^{-\beta}$ , the Hicksian area under the demand curve above the price line is  $\int_p^\infty kP^{-\beta} dp = \frac{k}{\beta-1} P^{1-\beta}$ , which equals  $\frac{PD}{\beta-1}$ . Summing over firms and segments gives the expression. The welfare comparison follows by algebra and the fact that coordination both lowers costs (network effects) and raises markups (price internalization).

**Sketch of proof (3.2.3).** Firm  $f$ 's objective is  $\sum_{mc} (P_{fmct}^{price} D_{fmct} - C_f^{oper})$ . Differentiate w.r.t  $P_{fmct}$  the revenue derivative is  $D_{fmct} + P_{fmct}^{price} \partial D_{fmct}/\partial P_{fmct}$ . Apply (71) for the cost term. Rearranging yields the stated First-Order Condition (FOC) and, with constant elasticity, the Lerner rule. If a price floor, cap, inter-period smoothness, or channel-parity constraint binds for some  $(f, m, c, t)$ , the first-order condition (73) is replaced by the corresponding KKT complementarity condition. In this case, the Lerner-type expression describes the unconstrained interior target, which is clipped by the active bound (Lerner, 1934). This does not affect existence or uniqueness of equilibrium but only determines whether the optimal price is interior or boundary-constrained.

**Sketch of proof (3.2.3).** Under constant-elasticity demand (25)–(26), the own-price term contributes curvature proportional to  $\beta_{mct}$ , while cross-price effects contribute off-diagonal terms proportional to  $\eta_{mct}^{ff'}$ . Binding shared constraints introduce additional coupling via common multipliers in the KKT system;  $\kappa_{mct}$  bounds the induced cross-firm effect on the VI Jacobian. Condition  $\beta_{mct} > \sum_{f' \neq f} \eta_{mct}^{ff'} + \kappa_{mct}$  yields uniform diagonal dominance of the symmetric Jacobian, implying strong monotonicity. Uniqueness then follows from standard VI theory on strongly monotone operators over convex compact sets. can be taken as any uniform bound on the magnitude of the additional off-diagonal terms in the symmetric Jacobian induced by common-dual coupling constraints; in particular  $\kappa_{mct} = 0$  when shared constraints are slack.

**Sketch of proof (3.2.3).** For each  $Y$ , Stage 1 has a unique equilibrium by Remark 5. The Stage 0 feasible set is a finite union of compact polyhedra; the objective is continuous in  $Y$  because  $P^*(Y)$  is continuous under a unique interior

equilibrium. Weierstrass' theorem guarantees an optimizer (Stone., 1948). Backward induction gives subgame-perfectness.

**Sketch of proof (3.2.3).** In Stage 0 the firms maximize  $\sum_{f \in \mathcal{F}} \Pi_f$  over  $Y$ ; choosing the “no-sharing” option  $Y = Y^{NC}$  reproduces the non-cooperative network, hence  $\sum_f \Pi_f^{HYB} \geq \sum_f \Pi_f^{NC}$ . The cooperative planner controls prices as well as  $Y$ , so it can replicate the hybrid outcome and weakly improve on it, giving the upper inequality.

**Sketch of proof (3.2.3).** Consider  $\phi(Y) = \sum_f \Pi_f(P^*(Y), Y)$ . By the envelope theorem for parameterized Nash equilibria,  $\partial \phi / \partial Q = \sum_f \partial \Pi_f / \partial Q$  because  $\partial \Pi_f / \partial P_f = 0$  at  $P^*$ ; the partial derivative equals the capacity dual at the lower level evaluated at equilibrium flows. Comparing the marginal present value  $\delta_t \psi_{k,t}^g$  to the average fixed-cost rate gives the activation test; if the shadow value is zero, there is no first-order gain.

#### 4.8. Full algorithm and pseudo-code

The following pseudo-code mirrors an implementation in JuMP/Pyomo with CPLEX/Gurobi (master and LP) and PATH/IPOPT (oracles). It integrates the feasibility logic, equilibrium solves, DCEC calibration, and stabilization; it is faithful to the narrative above.

Algorithm : LBB-EO with Dual-Calibrated Equilibrium Cuts (DCEC)

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1: Input data (sets, costs, capacities, distances, elasticities, taxes, bands);
2: Set termination controls: MaxIter,  $\varepsilon_{\text{gap}}$ ,  $\varepsilon_{\text{LP}}$ ,  $\varepsilon_{\text{MCP}}$ ,  $\Delta_{\text{rel}} \in [0.01, 0.05]$ ,  $\rho \in [1e-4, 1e-3]$ ;
3: Choose scenario SCEN  $\in \{\text{NC}, \text{HYB}, \text{COOP}\}$ ;  $\triangleright$  NC/HYB = Nash MCP; COOP = centralized concave NLP
4: Build  $\varepsilon$ -nearest sweeping sets  $N_j^\varepsilon$  and access rule  $C_j$ ; prune dominated arcs;
5: Define affine capacity map  $\kappa(Y)$  (prod/DC/green-DC/supplier/mode caps);
6: Initialize Master MILP over  $(Y \in \{0,1\}^n, \theta \in \mathbb{R})$  with budgets, logic (38)–(40), cost-share (46);  $\theta \leq \Theta_{\text{min}}$ ;
7: Initialize  $UB \leftarrow +\infty$ ,  $LB \leftarrow -\infty$ ,  $\Delta_H \leftarrow \Delta_0$ ;  $\text{CutPool} \leftarrow \emptyset$ ;  $it \leftarrow 0$ ;

8: while  $it < \text{MaxIter}$  and  $(UB - LB) / \max\{1, |UB|\} > \varepsilon_{\text{gap}}$  do
9:    $(Y^r, \theta^r) \leftarrow$  Solve Master MILP with  $\text{CutPool}$  and trust-region  $\|Y - \hat{Y}\|_1 \leq \Delta_H$ ;  $\triangleright \hat{Y} =$  current incumbent
10:   $LB \leftarrow \text{Objective}(\text{Master})$ ;  $\triangleright$  global lower bound

11:  # Operations oracle (exact LP/KKT)
12:  Build LP: minimize (1) subject to (2),(3)–(5),(9)–(11),(14),(16)–(17);  $\triangleright$  service proxy if endogenous
13:   $\text{status\_LP}, (z\_LP, \text{duals\_LP}) \leftarrow$  Solve LP to  $\varepsilon_{\text{LP}}$ ;
14:  if  $\text{status\_LP} = \text{infeasible}$  then
15:     $(\alpha, \beta) \leftarrow$  Farkas certificate projected to capacity space;  $\triangleright$  cover over  $\kappa(Y)$ 
16:    Add feasibility cut  $\alpha^T \kappa(Y) \geq \beta$  to  $\text{CutPool}$ ; goto line 27;

17:  # Price equilibrium oracle (scenario-specific)
18:  if SCEN = COOP then
19:    Solve coalition log-price NLP in  $s = \ln P$  with bands (33), smoothness (43), parity (44);
20:    Each evaluation solves Operations LP (lines 12–13); stop at KKT residual  $\leq \varepsilon_{\text{LP}}$ ;
21:  else # SCEN  $\in \{\text{NC}, \text{HYB}\}$ 
22:    Form MCP: price-KKT for both firms (with (33),(43),(44)) + operations-KKT; add small proximal  $\rho \|s - s_{\text{prev}}\|^2/2$ ;
23:    Solve MCP with PATH to natural residual  $\leq \varepsilon_{\text{MCP}}$ ;
24:  end if
25:  Extract  $\Phi(Y^r)$ ,  $P^*$ , flows  $z^*$ , and capacity duals  $\lambda_{\text{raw}} \in \{\xi, \gamma, \gamma^g, \sigma, \psi\}$  (at equilibrium);
26:   $UB \leftarrow \min\{UB, \Phi(Y^r) - \text{FixedActivation}(Y^r)\}$ ; update incumbent  $(\hat{Y}, \hat{P}, \hat{z}, \hat{\lambda})$  if improved;

27:  # Dual-Calibrated Equilibrium Cut (DCEC)
28:   $c^* \leftarrow \arg\max_c |\lambda_{\text{raw}, c}|$ ;  $\Delta_c \leftarrow \max\{1, \Delta_{\text{rel}} \cdot \kappa_c(Y^r)\}$ ;  $\tilde{\kappa} \leftarrow \kappa(Y^r)$ ;  $\tilde{\kappa}_c^* \leftarrow \tilde{\kappa}_c^* + \Delta_c$ ;
29:  if SCEN = COOP then
30:     $\Phi_\Delta \leftarrow$  Re-solve coalition log-price NLP at capacity  $\tilde{\kappa}$ ;  $\triangleright$  same tolerances
31:  else
32:     $\Phi_\Delta \leftarrow$  Re-solve MCP at  $\tilde{\kappa}$  with proximal weight  $\rho$  in  $\ln$ -prices;  $\triangleright$  keep active set stable
33:  end if
34:   $\lambda_{\text{cal}} \leftarrow \lambda_{\text{raw}}$ ;  $\lambda_{\text{cal}, c^*} \leftarrow (\Phi_\Delta - \Phi(Y^r)) / \Delta_c$ ;  $\triangleright$  one-sided secant slope

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35: Add optimality cut  $\theta \leq \Phi(Y^r) + \langle \lambda_{\text{cal}}, \kappa(Y) - \kappa(Y^r) \rangle$  to CutPool;
36: Update trust-region radius  $\Delta_H$  (expand if last cut strongly violated; shrink on oscillations);

37: it  $\leftarrow$  it + 1;
38: end while

39: Return  $Y^*$ ,  $P^*$ ,  $z^*$ , duals* from incumbent; also report:
40: • Master gap  $(UB - LB)/\max\{1, |UB|\} \leq \varepsilon_{\text{gap}}$ ,
41: • LP primal/dual infeasibilities  $\leq \varepsilon_{\text{LP}}$ , complementarity sum  $\leq \varepsilon_{\text{LP}}$ ,
42: • MCP natural residual  $(NC/HYB) \leq \varepsilon$  MCP (or NLP KKT residual in COOP).

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**Figure 2.** Pseudo-code of our proposed LBB–EO–DCEC.

What the pseudo-code is doing in mathematical terms. Each master iteration proposes a design  $Y^r$  and a lower bound. The LP oracle either returns a Farkas feasibility certificate, which becomes a linear cover inequality over  $\kappa(Y)$ , or it yields primal flows and duals. The price oracle returns an equilibrium consistent with (NC/HYB) MCP or COOP concave log-price program. The algorithm then forms an optimality cut that is linear in  $Y$  because capacities are affine in  $Y$ . When prices are at corners, the DCEC replaces the raw dual slope by a secant computed from two exact equilibrium values, ensuring validity and improving tightness. Because the design space is finite and each revisit yields a new feasibility or optimality cut, the loop terminates finitely once the upper–lower gap is within tolerance. The returned solution is a structure  $Y^*$  with true equilibrium prices  $P^*$  and flows that satisfy all constraints, together with certificates (LP residuals, MCP residual, master gap) that reviewers can use to verify optimality and feasibility.

#### 4.9. Relationship to Standard Logic-Based Benders Decomposition (LBBD)

The proposed solution framework builds on the logic-based Benders decomposition (LBBD) paradigm but departs from the classical formulation in several fundamental and necessary ways to accommodate strategic pricing equilibria, non-smooth service effects, and carbon-regulated competition.

In standard LBBD, the original problem is decomposed into (i) a master problem that optimizes over discrete design or policy decisions and (ii) one or more continuous subproblems, typically linear or convex nonlinear programs. Given a fixed master design, the subproblem either returns an optimal objective value together with valid dual multipliers used to generate Benders optimality cuts or produces an infeasibility certificate (e.g., a Farkas ray) that yields feasibility cuts. This classical structure relies critically on two assumptions: (i) the subproblem is a well-defined convex optimization problem, and (ii) its dual multipliers provide globally valid marginal values for constructing supporting cuts.

While our framework shares the high-level master–subproblem separation of LBBD, these assumptions do not hold in the present setting. Here, feasibility and optimality are not determined by the operational LP alone. Instead, the realized objective value depends on an equilibrium-consistent pricing layer, which fundamentally alters the nature of the “subproblem.” Specifically, conditional on a candidate strategic design  $Y^r$ , the lower layer consists of two tightly coupled components: (i) an exact operational LP that determines production, inventory, multimodal logistics, service levels, and emissions, and (ii) a price equilibrium problem that governs market outcomes. In the NC and HYB regimes, prices arise from a generalized Nash equilibrium formulated as a monotone MCP. In the COOP regime, prices result from a centralized but constrained concave pricing optimization. Consequently, the “subproblem” is not a single convex program but an equilibrium oracle whose solution must simultaneously satisfy operational feasibility and equilibrium optimality conditions.

This distinction is crucial. Although LP dual multipliers correctly price marginal changes in capacities and flows, they may misrepresent the true marginal impact on the equilibrium objective when prices are restricted by policy bands, parity constraints, inter-temporal smoothness limits, or when service levels are governed by non-smooth operators such as min or saturation functions. Near such kinks, classical Benders optimality cuts constructed from raw duals tend to be weak or misleading, allowing the master to propose designs that appear profitable in relaxation but are not

attainable under equilibrium behavior. As a result, a direct application of standard LBBD may converge slowly or oscillate. To address these limitations, we extend the LBBD framework along two key methodological dimensions.

*First, we replace conventional subproblem solvers with Equilibrium Oracles (EO).*

Given a candidate design  $Y^r$ , the oracle computes a fully consistent equilibrium outcome prices, flows, service levels, and emissions by solving either (i) a Nash MCP (NC/HYB) or (ii) a concave coalition pricing problem (COOP), embedding the exact operational LP. Feasibility is therefore defined not only by operational constraints but also by equilibrium conditions. This guarantees that every incumbent solution, reported bound, and oracle evaluation is economically meaningful and equilibrium-consistent.

*Second, we introduce Dual-Calibrated Equilibrium Cuts (DCEC).*

Rather than relying solely on raw dual multipliers from the operational LP, DCEC locally calibrate marginal values using one-sided secant slopes of the equilibrium value function with respect to selected components of the affine capacity map  $\kappa(Y)$ . These slopes are computed by applying a small, equilibrium-preserving perturbation to the design, re-solving the equilibrium oracle, and measuring the resulting change in the equilibrium objective. Because DCEC are constructed from exact oracle evaluations, they remain valid lower-bounding cuts while being substantially tighter than classical Benders tangents in the presence of non-smooth price or service effects.

From a methodological standpoint, the proposed LBB–EO–DCEC framework can therefore be viewed as a generalization of logic-based Benders decomposition to equilibrium-constrained design problems. Compared with standard LBBD, it (i) replaces convex subproblems with equilibrium oracles, (ii) augments dual-based cuts with calibrated equilibrium slopes, and (iii) preserves a pure MILP master with certified upper and lower bounds. Compared with monolithic MPCC or EPEC reformulations, it avoids large nonconvex mixed-integer programs and instead achieves numerically stable convergence through decomposition, with explicit feasibility and optimality certificates.

In summary, the proposed approach retains the conceptual simplicity of LBBD a master–oracle architecture with iterative cut generation while extending it in a principled way to handle strategic pricing, non-smooth service effects, and carbon-regulated competition that a standard decomposition cannot capture. This positioning clarifies both the rigor and the practical value of the method for large-scale, equilibrium-driven supply chain design problems.

Appendix AA. Literature Synthesis and Comparative Positioning

Table A1. Tabular summary of the literature (pricing–inventory–carbon–marketing–games)

| Study                   | Supply chain setting                                                                       | Core decisions                                                                     | Carbon / policy instrument                                       | Demand / behavior features                                                   | Modeling / solution approach                                          | Main takeaways (managerial + theoretical)                                                                                                                                                                                                             |
|-------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Choudhury et al. (2025) | Retailer-centric (multi-product categories; supplier offers volume discount + trade terms) | Retail price, advertising, service investment, cycle time                          | Cap-and-price, cap-and-trade, carbon tax                         | Demand influenced by price + promotion + service                             | Analytical structure + novel algorithm, numerical + sensitivity       | Jointly optimizing marketing + operational levers can improve profit and emissions performance; regulation choice matters—policy regimes change the “best” operating region and investment intensity.                                                 |
| Sun et al. (2025)       | 1 supplier – 1 manufacturer                                                                | Low-carbon R&D (abatement level), strategic mode choice                            | Carbon tax (SDG framing)                                         | Market/environment performance metrics (profit, consumer surplus, emissions) | Equilibrium comparisons across cooperative vs independent R&D modes   | A sufficiently high tax can stimulate joint abatement when R&D cost coefficient is low; “simultaneous independent R&D” can dominate single-firm R&D in abatement and welfare; cooperation only-in-R&D can yield maximal abatement in their extension. |
| Yan & Ni (2024)         | Manufacturer–retailer                                                                      | Selling prices + emission reduction rates                                          | Cap-and-trade                                                    | Uncertainty theory: parameters as uncertain variables (expert belief)        | Decentralized game models; analytical solutions + numerical           | Under limited data and non-frequency-stable parameters, uncertain-game modeling produces tractable equilibria and useful comparative insights across decentralized structures.                                                                        |
| Yadav et al. (2024)     | Deteriorating inventory system (single firm / retailer view)                               | Selling price, preservation technology, carbon reduction investment                | Emission reduction measures (regulatory context)                 | Price-dependent demand; deterioration slowed by PT; waste/salvage            | Inventory model + numerical/sensitivity                               | Preservation investment tied to price can be profit-improving and emission-reducing; highlights role of PT + abatement as joint levers, with salvage handling.                                                                                        |
| Mahato et al. (2024)    | Sustainable inventory with controllable deterioration + trade credit                       | Price, preservation investment, cycle time                                         | Multiple regimes: cap-and-price, carbon tax, cap-and-trade, none | Deterioration controllable; financing via trade credit                       | Theoretical properties + algorithm + examples/sensitivity             | Carbon policy ranking can differ: in their comparisons, cap-and-price can yield higher profit than other regimes; trade credit interacts with sustainability levers and deterioration control.                                                        |
| Zhang & Yu (2023)       | Low-carbon closed-loop SC: manufacturer–retailer–government                                | Pricing, emission-reduction investment, recycling investment; government subsidies | Government double subsidy + dynamic setting                      | Goodwill and recycling rate dynamics                                         | Three-level differential games; equilibrium derivations + sensitivity | Power structure shapes subsidy rates and private investments; weaker party tends to receive higher subsidy; non-led structure may improve recycling/social welfare under subsidies; contracts can coordinate.                                         |

|                                |                                                       |                                                                         |                                                 |                                                                |                                                                              |                                                                                                                                                                                                             |
|--------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Li & Mizuno (2022)             | Dual-channel manufacturer–retailer                    | Dynamic pricing + inventory control                                     | None (baseline RM / DP)                         | Stochastic price-sensitive demand                              | Stochastic dynamic programming; power structure comparisons                  | Optimal policy structure resembles base-stock + list-price type; power structure changes levels (not structural form); leadership affects distribution of benefits and decisions.                           |
| Pourmohammad-Zia et al. (2021) | Food SC with growing + deteriorating items            | Dynamic pricing + inventory; coordination contract                      | None                                            | Growth function + deterioration; waste considerations          | Nonlinear convex optimization; centralized vs decentralized + profit-sharing | Coordination can reduce prices and increase total profit; shows how biology/growth + deterioration reshape dynamic pricing and replenishment timing.                                                        |
| Sheng et al. (2023)            | Pricing strategy selection                            | Dynamic pricing vs pre-announced pricing                                | None                                            | Informed/uninformed + patience heterogeneity                   | 2-period game-theoretic analysis                                             | Which pricing regime dominates depends on consumer mix and patience; decentralized setting can still yield “win-win” regions for firms + consumers.                                                         |
| Ranjan & Jha (2022)            | 2-echelon deteriorating product SC                    | Multi-period dynamic retail pricing; replenishment cycle; PT investment | Carbon emission included                        | Demand depends on price, inventory, reference price, freshness | Algorithmic optimization + numerical/sensitivity                             | PTI + dynamic pricing can strongly raise profits; reference price is a key driver; carbon costs shift the PTI–pricing trade-off frontier.                                                                   |
| Parilina et al. (2025)         | Manufacturer–retailer (Stackelberg differential game) | Dynamic pricing + advertising (and contract choice)                     | “Environment- and fairness-concerned consumers” | Reference price fairness + green info sensitivity              | Differential game; compare wholesale vs revenue-sharing                      | Contract form changes whether outcomes can be Pareto-improving (members + consumers + environment); fairness/green disclosure reshapes optimal pricing/advertising paths.                                   |
| Zarouri et al. (2022)          | Perishable product SC                                 | Two-period markdown pricing; cooperation vs Stackelberg                 | None                                            | Stochastic, quality-dependent demand                           | Heuristic analysis of games                                                  | Cooperation yields lowest prices and highest total profit; leadership changes price markdown intensity; highlights coordination value under perishability.                                                  |
| Yan et al. (2024)              | Manufacturer + digital retailer + B&M retailer        | Dynamic pricing + emission reduction + promotions                       | Nonexplicit (green reputation)                  | Bidirectional free-riding, reputation dynamics                 | Tripartite dynamic game                                                      | Free-riding fundamentally changes how promotions and reputation translate into prices/profit; reputation effects can flip depending on cross-effects; parameters (decay, cost coefficients) define regimes. |
| Rahmani & Pashapour (2024)     | Dual channel with new + returned products             | Prices for new/refurbished + network decisions (bi-objective MINLP)     | Sustainability framing                          | Customer segmentation: return behavior modeled                 | Bi-objective MINLP + metaheuristics                                          | More behavioral return modeling improves realism; dynamic prices across segments can improve profit while meeting social objective.                                                                         |

|                         |                                                      |                                                                 |                                        |                                                                     |                                                                                     |                                                                                                                                                                          |
|-------------------------|------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shu et al. (2025)       | Two-period CLSC with collection inconvenience        | Pricing + collection investment                                 | Nonexplicit                            | Channel inconvenience; competition in collection                    | Game-theoretic + numerical                                                          | Collection competition improves environmental performance but hurts profits and welfare; coordination contract (two-part tariff) can restore performance.                |
| Alamdar & Seifi (2024)  | Retailer with strategic customers; seasonal products | Ordering + dynamic pricing                                      | None                                   | Strategic waiting; multiple substitutes                             | Deep Q-learning (RL) + real dataset                                                 | RL can outperform classic metaheuristics in large state spaces; demand learning + pricing/ordering integration is powerful when closed-form optimization is hard.        |
| Bai et al. (2024)       | Advance selling policies                             | Pricing + ordering under advance selling variants + advertising | None                                   | Stochastic valuations: advertising affects diffusion                | Analytical solutions + numerical                                                    | The best policy depends on share of high-valuation customers; advertising changes diffusion and profitability; identifies thresholds for when advance selling dominates. |
| Zhou et al. (2022)      | On-site services (routing + pricing)                 | Vehicle routing (CO <sub>2</sub> ) + time-window pricing        | Emission reduction objective           | Customers choose time window based on price                         | Two-phase: low-emission VRP then pricing optimization; metaheuristic for online use | Dynamic pricing can shift demand to cleaner schedules and raise profit; useful template for coupling “emissions-minimizing ops” with “profit-maximizing pricing.”        |
| Schlosser et al. (2021) | Circular economy (firm-level control)                | Dynamic pricing + recycling investment                          | Circularity objective (recycling rate) | Recycling affects supply + demand, diminishing effect over time     | Optimal control; steady state + sensitivity                                         | Joint control of price and recycling can stabilize sustainability trajectories; investment combats the natural decay of “green demand response.”                         |
| Tunçınan et al. (2024)  | 1 warehouse – many heterogeneous retailers           | Allocation + pricing over horizon                               | None                                   | Deterministic price-sensitive demand; lost sales                    | MIQCP + structural results + heuristic                                              | When inventory is large, tractability improves; retailer heterogeneity & seasonality drive optimal price/allocation paths; decomposition heuristic yields small gaps.    |
| Soares et al. (2020)    | Electricity retail tariffs                           | Bi-level TOU pricing                                            | None                                   | Consumer response in lower level; lower-level hard to solve exactly | Bi-level + bound estimation under computation budget                                | When LL can’t be solved exactly, bounding becomes essential for reliable leader decisions; a strong methodological reference for BL settings.                            |
| Gholami et al. (2021)   | Bi-level channel optimization, multi-period          | Dynamic pricing with stochastic demand memory                   | None                                   | Price-dependent demand with memory                                  | Iterative algorithm that decouples nested equilibria                                | Demand “memory” makes equilibria nested; iterative decoupling gives explicit equilibria across broad scenarios/time horizons.                                            |
| Song et al. (2021)      | Seller–buyer dynamic Stackelberg                     | Wholesale price + innovation efforts over time                  | None                                   | Goodwill dynamics: innovation affects demand and cost               | Differential Stackelberg game                                                       | Innovation + pricing coevolve; leadership changes profit and goodwill paths; useful for “dynamic effort + price” modeling.                                               |

|                          |                                              |                                                           |                                 |                                                               |                                                            |                                                                                                                                                                       |
|--------------------------|----------------------------------------------|-----------------------------------------------------------|---------------------------------|---------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pazoki et al. (2024)     | Regulated transport capacity (Canadian case) | Equilibrium pricing + quota regulation                    | Government intervention (quota) | Disruption-driven capacity scarcity                           | Market models + algorithm                                  | Justifies when intervention is necessary; quota stabilizes outcomes under capacity scarcity; bridges resilience + regulation + equilibrium pricing.                   |
| Wang & Du (2025)         | Incentive programs                           | Price discrimination + quality discrimination             | Data compliance policy framing  | Customer incentive programs + patience                        | Analytical comparisons                                     | Firms may offer lower prices + higher quality to attract new customers; rewarding old customers can be suboptimal; policy constraints affect welfare.                 |
| Dye (2020)               | Perishable items; multi-period               | Dynamic pricing + advertising + inventory (psychic stock) | None                            | Demand depends on price, freshness, goodwill, display stock   | Infinite-horizon dynamic optimization                      | Under some settings, optimal pricing can become static while goodwill/stock dynamics converge; gives structural properties for DP-advertising-inventory coupling.     |
| Qiu et al. (2023)        | Retailer with quick response                 | 2-period pricing + ordering                               | None                            | Distributionally robust market size uncertainty               | DRO reformulations; closed-form conditions                 | Dynamic pricing benefits depend on uncertainty and quick response; DRO gives tractable robustness when distribution is unknown.                                       |
| Zhou et al. (2023)       | Perishable, partial lost sales               | Pricing + two-stage inventory (reorder/return)            | None                            | Random demand; old vs fresh competition                       | Dynamic programming                                        | Mid-season reorder decreases with leftover quantity/quality; shows how “quality-differentiated inventory” shapes pricing and replenishment.                           |
| Dye & Hsieh (2024)       | Green innovation + pricing + advertising     | Dynamic pricing + green innovation + advertising          | Cap-and-price                   | Demand multiplicative in price and goodwill; goodwill evolves | Infinite-horizon dynamic optimization + structural results | Green innovation and advertising build goodwill; cap-and-price changes steady-state and policy sensitivities; provides interpretable structural effects.              |
| Crettez et al. (2025)    | Competing supply chains with coordination    | Capacity choice + retail price under revenue-sharing      | None                            | Complementary goods; discontinuity issues                     | Differential game; equilibrium existence analysis          | Revenue-sharing can destroy equilibrium existence due to discontinuities—important caution for “coordination contracts + competition” modeling.                       |
| Hamidoğlu & Weber (2024) | Low-carbon agriculture; collaborative market | Low-carbon adoption efforts + collaborative pricing       | Subsidies + carbon taxes        | Cooperative Nash interactions                                 | Nash-based operational framework + case studies            | Proper subsidy/tax design stabilizes low-carbon transition; collaboration can align adoption and affordability; policy can reduce cost gap vs conventional.           |
| Han et al. (2025)        | Dyadic SC with fairness concerns             | Pricing under power structures                            | None                            | Fairness concerns affect decisions                            | Stackelberg vs Nash comparisons                            | Fairness of leaders increases retail prices; power balance changes profit distribution and total performance; fairness expands pricing theory under power structures. |

**Table A2.** Comparative positioning of this study vs. state of the art

| Dimension                     | Representative state-of-the-art studies                                         | What existing studies do                                          | Key limitation in literature                                         | This study (MEMM-SCN)                                                           |
|-------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Supply chain scope            | Li & Mizuno (2022); Sun et al. (2025); Yan & Ni (2024)                          | Focus on single- or two-echelon supply chains; usually one market | Cannot capture cross-echelon and cross-market spillovers             | Multi-echelon, multi-market, multi-channel network with endogenous flows        |
| Pricing structure             | Li & Mizuno (2022); Sheng et al. (2023); Wang et al. (2024)                     | Dynamic pricing, typically single-channel or dual-channel         | Pricing is not linked to network topology or logistics decisions     | Channel-specific dynamic pricing jointly optimized with network configuration   |
| Carbon regulation modeling    | Choudhury et al. (2025); Sun et al. (2025); Yan & Ni (2024); Dye & Hsieh (2024) | Carbon tax, cap-and-trade, or cap-and-price studied in isolation  | Carbon policy effects analyzed without endogenous logistics pooling  | Multiple carbon regimes embedded directly in firm-level constraints             |
| Sustainability levers         | Yadav et al. (2024); Ranjan & Jha (2022); Schlosser et al. (2021)               | Investment in abatement, preservation, or recycling               | Sustainability decisions treated separately from pricing competition | Carbon mitigation + service + pricing decisions co-optimized                    |
| Inter-firm behavior           | Sun et al. (2025); Zhang & Yu (2023); Shu et al. (2025)                         | Either pure competition or full cooperation                       | Binary behavioral assumption unrealistic in practice                 | Three regimes: NC, COOP, and novel HYB coopetition                              |
| Nature of cooperation         | Zhang & Yu (2023); Yi et al. (2024)                                             | Cooperation via contracts or centralized control                  | No partial cooperation preserving price competition                  | Shared logistics & green assets + price competition (coopetition)               |
| Game-theoretic structure      | Li & Mizuno (2022); Zhang et al. (2025 IEEE TNSE)                               | Nash or Stackelberg games, usually separable                      | Cannot handle shared constraints across firms                        | Generalized Nash Equilibrium (GNE) with shared logistics and carbon constraints |
| Optimization hierarchy        | Soares et al. (2020); Gholami et al. (2021)                                     | Bi-level models focused on pricing or tariffs                     | Network design rarely appears at strategic level                     | Bi-level: strategic GNE (upper) + operational profit maximization (lower)       |
| Mathematical complexity       | Tunçinan et al. (2024); Rahmani & Pashapour (2024)                              | MILP / MINLP without equilibrium consistency                      | Discrete network + equilibrium often approximated                    | Multi-level MINLP with equilibrium constraints                                  |
| Solution methodology          | Classical Benders; heuristics; metaheuristics (Alamdard & Seifi, 2024)          | Heuristics lack equilibrium guarantees                            | No equilibrium-consistent decomposition                              | Logic-Based Benders with Equilibrium Oracles (LBB-EO) + DCEC                    |
| Logistics–pricing interaction | Zhou et al. (2022); Pazoki et al. (2024)                                        | Pricing affects routing or quotas exogenously                     | No endogenous topology reconfiguration                               | Pricing reshapes topology and modal split (truck → rail)                        |
| Resilience under constraints  | Pazoki et al. (2024); Zhang & Yu (2023)                                         | Focus on single disruption or policy stress                       | Strategic regime robustness unexplored                               | COOP shown most robust under rail scarcity & binding quotas                     |

|                          |                                                 |                                        |                                    |                                                                         |
|--------------------------|-------------------------------------------------|----------------------------------------|------------------------------------|-------------------------------------------------------------------------|
| Empirical grounding      | Many studies rely on stylized or synthetic data | Limited industrial realism             | Results are hard to operationalize | Real electronics supply chain case with empirical demand, carbon prices |
| Managerial insight level | Choudhury et al. (2025); Mahato et al. (2024)   | Pricing- or inventory-centric insights | Cannot explain structural shifts   | End-to-end insights: topology, emissions, profitability, resilience     |

## Appendix BB. KKT Implementation of the Lower-Level Model and Single-Level Reformulations

### B.1. Primal LP in standard form (given strategic binaries and prices)

Fix a scenario (NC/COOP/HYB), a strategic activation/investment plan (plants, DCs, green DCs, awards, sweeping assignments), and a posted price vector  $P$ . The lower level minimizes augmented operating cost (production, transport, inventory, and any carbon tax/permit cost pushed into unit costs per section 3.1), subject to feasibility. Stack all nonnegative continuous lower-level variables in  $z$  (these include all flows  $x$ , production  $q$ , inventories  $I$ , and if the service proxy is embedded auxiliaries  $N^v, T, S^v, s$  see section A.5). Write the LP as

$$\min_{z \geq 0} C^\top z \quad \text{s.t.} \quad Az = b(P, \ell), \quad Gz \leq h \quad (\text{KKT.1})$$

Where  $Az = b$ , collects equalities (plant outbound balance (16), DC and green-DC inventory dynamics (9)–(10), market clearing (11), definitions for  $N^v, T, S^v$ , sif used), and  $Gz \leq h$  collects capacities and gating ((2)–(5), (14), (17), sweeping and award linkages from (59)–(61) and (31)–(32)). The right-hand side  $b(P, \ell)$  contains the demand  $D_{fmc}^v(P, \ell)$  in (11). if  $\ell$  is endogenized we add its defining constraints ((19)–(23)) to  $Az = b$  as in section A.5. Unit-cost vector  $c$  is already augmented for carbon per section 3.1.2 (so the KKT remains linear). Indexing and arc categories used below.

Supplier→plant  $x_{ifjat}^v$ ; plant→private-DC  $x_{fjkpt}^v$ ; plant→green-DC  $x_{fjk'pt}^v$ ; private-DC→market  $x_{fkmcpt}^v$ ; green-DC→market  $x_{fk'mcpt}^v$ . Production  $q_{fjpt}^{\text{prod}}$ ; private-DC inventory  $I_{fjkpt}$ ; green-DC inventory  $I_{fk'pt}$ . Sweeping assignment binaries  $w_{ijat}$  and award binaries  $y_{iat}^{\text{sup}}$  are upper level (fixed here).

### B.2. Dual, KKT, and economic multipliers

Let  $y$  be the free dual vector for equalities  $Az = b$  and  $u \geq 0$  the multipliers for capacities  $Gz \leq h$ . The dual is

$$\min_{y \text{ free}, u \geq 0} b(P, \ell)^\top y + h^\top u \quad \text{s.t.} \quad A^\top y + G^\top u \leq c \quad (\text{KKT.2})$$

The KKT system is:

- *Primal feasibility*:  $Az = b(P, \ell), Gz \leq h, z \geq 0$ .
- *Dual feasibility*:  $A^\top y + G^\top u \leq c, u \geq 0$ .
- *Complementary slackness (pairwise)*: for each capacity row  $g$ ,  $u_g \perp (h_g - g^\top z)$ ; for each nonnegative variable  $z_i$ ,  $z_i \perp r_i$ , where the reduced cost

$$r = c - A^\top y + G^\top u \text{ satisfies } r \geq 0 \quad (\text{KKT.3})$$

We name key duals for interpretation:

- $\phi_{fjpt}$ (free): plant outbound balance (16).
- $\eta_{fjkpt}$  and  $\eta_{fk'pt}^g$ (free): private and green DC inventory dynamics (9)–(10).
- $\mu_{fmc}$ (free): market clearing (11) (this is the “all-inclusive marginal cost” used in price FOCs).
- $\xi_{fjpt} \geq 0$ : production capacity (2).
- $\gamma_{fjkpt}^{\text{in}}, \gamma_{fjkpt}^{\text{out}} \geq 0$ : private-DC inbound/outbound (4)–(3).
- $\gamma_{k't}^g \geq 0$ : green-DC shared capacity (5).
- $\sigma_{iat} \geq 0$ : supplier capacity (14).
- $\psi_{vt} \geq 0$ : mode capacity (17).
- $\pi_{iat} \geq 0$ : award-gate (sum over  $f, j, v$  flows  $\leq \bar{M}_{at} y_{iat}^{\text{sup}}$ ).

- $\varpi_{ijat} \geq 0$ : sweeping-gate  $x_{ifjat}^v \leq \bar{M}_{jat} w_{ijat}$ .

(If emissions are not pushed into  $c$ , include an emission accounting equality with dual  $\kappa_t$ ; see section A.6.)

### B.3. Stationarity (reduced-cost) equations by variable family

The reduced-cost form  $r = c - A^\top y - G^\top u$  makes signs unambiguous. For any variable  $z_i > 0$ , its reduced cost  $r_i = 0$ ; otherwise  $r_i \geq 0$ .

Flows from supplier to plant  $x_{ifjat}^v$ , This variable appears in, plant material balance (6) on the left with +1 coefficient (we write it as an equality by adding a nonnegative slack for over-supply), supplier capacity (14) with +1, mode capacity (17) with +1, sweeping gate with +1, and award gate with +1. It does not enter DC or market equalities directly. Its unit cost is  $c_{v,i \rightarrow j,t}^{\text{Tran}}$  (already carbon-augmented). Hence

$$r_{i,f,j,a,t}^{\text{sup} \rightarrow \text{pl}} = c_{v,i \rightarrow j,t}^{\text{Tran}} - \alpha_{fjat} - \sigma_{iat} - \psi_{vt} - \varpi_{ijat} - \pi_{iat} \geq 0, \quad x_{ifjat}^v \perp r_{i,f,j,a,t}^{\text{sup} \rightarrow \text{pl}} \quad (\text{KKT.4})$$

where  $\alpha_{fjat}$  is the free dual of the material balance equality  $\sum_{i,v} x_{ifjat}^v - \sum_p r_{ap} q_{fjpt}^{\text{prod}} - s_{fjat}^{\text{mat}} = 0$  (we introduce  $s_{fjat}^{\text{mat}} \geq 0$  to convert (6) into an equality).

Production  $q_{fjpt}^{\text{prod}}$ , This variable appears in, plant outbound balance (16) with -1, production capacity (2) with +1, material balance equalities with  $-r_{\{a p\}}$ , and (optionally) emission accounting with  $+e_{fjp}^{\text{prod}}$  if not pushed into cost. Unit cost is  $c_{fjpt}^{\text{prod}}$  (carbon-augmented). Then

$$r_{fjpt}^{\text{prod}} = c_{fjpt}^{\text{prod}} + \xi_{fjpt} - \sum_p r_{ap} \alpha_{fjat} - \underbrace{\kappa_t e_{fjp}^{\text{prod}}}_{\text{Omit if embedded}} + \dots + (\phi_{fjpt}), \quad (\text{KKT.5})$$

but careful with the sign of  $\phi$ : since (16) is  $\sum_{k,v} x_{fjkpt}^v + \sum_{k',v} x_{fjk'pt}^v - q_{fjpt}^{\text{prod}} = 0$ , the coefficient on  $q$  is -1, so  $r = c - A^\top y - \dots$  contributes  $-(-1)\phi = +\phi$ . Hence

$$r_{fjpt}^{\text{prod}} \geq 0, \quad q_{fjpt}^{\text{prod}} \perp r_{fjpt}^{\text{prod}} \quad (\text{KKT.6})$$

Flows from plant to private DC  $x_{fjkpt}^v$ , This variable appears in, plant outbound balance (16) with +1, private - DC inventory equality (9) with +1 (we write inventory as  $(1 - \phi_k)I_{t-1} + \sum_{j,v} x_{fjkpt}^v - \sum_{m,c,v} x_{fkmcpt}^v - I_{fkpt} = 0$ ), inbound DC capacity (4) with +1, and mode capacity (17) with +1. Unit cost  $c_{v,j \rightarrow k,t}^{\text{Tran}}$  (carbon - augmented). Then

$$r_{fjkpt}^{\text{pl} \rightarrow \text{pdc}} = c_{v,j \rightarrow k,t}^{\text{Tran}} - \phi_{fjpt} - \eta_{fkpt} - \gamma_{fkpt}^{\text{in}} - \psi_{vt} \geq 0, \quad x_{fjkpt}^v \perp r_{fjkpt}^{\text{pl} \rightarrow \text{pdc}} \quad (\text{KKT.7})$$

Flows from plant to green DC  $x_{fjk'pt}^v$ , Analogous to private DC  $x_{fjkpt}^v$  but using the green-DC inventory dual  $\eta_{fk'pt}^g$ :

$$r_{fjk'pt}^{\text{pl} \rightarrow \text{gdc}} = c_{v,j \rightarrow k',t}^{\text{Tran}} - \phi_{fjpt} - \eta_{fk'pt}^g - \psi_{vt} \geq 0, \quad x_{fjk'pt}^v \perp r_{fjk'pt}^{\text{pl} \rightarrow \text{gdc}} \quad (\text{KKT.8})$$

Flows from private DC to market  $x_{fkmcpt}^v$ , This variable enters, private-DC inventory equality (9) with -1, market-clearing equality (11) with +1, outbound DC capacity (3) with +1, mode capacity (17) with +1. Unit cost  $c_{v,k \rightarrow (m,c),t}^{\text{Tran}}$ . Hence

$$r_{fkmcpt}^{\text{pdc} \rightarrow \text{mkt}} = c_{v,k \rightarrow (m,c),t}^{\text{Tran}} + \eta_{fkpt} - \mu_{fmct} - \gamma_{fkpt}^{\text{out}} - \psi_{vt} \geq 0, \quad x_{fkmcpt}^v \perp r_{fkmcpt}^{\text{pdc} \rightarrow \text{mkt}} \quad (\text{KKT.9})$$

Flows from green DC to market  $x_{fk'mcpt}^v$ , Use green-DC inventory dual  $\eta^g$  and shared capacity dual  $\gamma^g$ :

$$r_{fk'mcpt}^{gdc \rightarrow mkt} = c_{v,k' \rightarrow (m,c),t}^{Tran} + \eta_{fk'pt}^g - \mu_{fmct} - \gamma_{k't}^g - \psi_{vt} \geq 0, \quad x_{fk'mcpt}^v \perp r_{fk'mcpt}^{gdc \rightarrow mkt} \quad (\text{KKT.10})$$

Inventories  $I_{fkpt}$  and  $I_{fk'pt}$ , For private DCs, the inventory equality at  $t$  is  $(1 - \phi_k)I_{fkpt,t-1} + \text{inflow}_t - \text{outflow}_t - I_{fkpt} = 0$  with dual  $\eta_{fkpt}$ .  $I_{fkpt}$  also appears in period  $t + 1$  with coefficient  $+(1 - \phi_k)$ . With unit holding cost  $c_{fkpt}^{Inv}$ ,

$$r_{fkpt}^{Inv} = c_{fkpt}^{Inv} - \eta_{fkpt} - (1 - \phi_k)\eta_{fkpt,t+1} \geq 0, \quad I_{fkpt} \perp r_{fkpt}^{Inv} \quad (\text{KKT.11})$$

with the terminal condition  $\eta_{fkpt,T+1} \equiv 0$ . The green-DC case is identical with  $\eta^g$  and  $c_{fk'pt}^{Inv}$ .

Material-balance slack  $s_{fjat}^{\text{mat}}$ , Arises when we convert (6) to equality. Since it enters with coefficient  $-1$  and has zero-unit cost,

$$r_{fjat}^{\text{mat-slack}} = 0 - (-\alpha_{fjat}) = +\alpha_{fjat} \geq 0, \quad s_{fjat}^{\text{mat}} \perp \alpha_{fjat} \quad (\text{KKT.12})$$

At optimum we typically get  $s^{\text{mat}} = 0$  and  $\alpha_{fjat} \leq 0$  (non-binding material balance), but writing the equality keeps the KKT system clean.

#### B.4. Complementary-slackness pairs for capacities

For every inequality row “ $\leq$ ”, include the orthogonality  $u_g \perp (h_g - g^T z)$ . Explicitly:

- *Production capacity* (2):  $\xi_{fjpt} \perp (\bar{Q}_{fjpt}^{\text{prod}} + U_{fjp}^{\text{cap}} \Delta \bar{Q}_{fjpt}^{\text{prod}}) y_{fjpt}^{\text{PL}} - q_{fjpt}^{\text{prod}}$ .
- *Private-DC inbound/outbound* (4)/(3):  $\gamma_{fkpt}^{\text{in}} \perp \bar{Q}_{fkpt}^{\text{DC}} y_{fkpt}^{\text{DC}} - \sum_{j,v} x_{fjkpt}^v$ ; and  $\gamma_{fkpt}^{\text{out}} \perp \bar{Q}_{fkpt}^{\text{DC}} y_{fkpt}^{\text{DC}} - \sum_{m,c,v} x_{fkmcpt}^v$ .
- *Green-DC shared* (5):  $\gamma_{k't}^g \perp \bar{Q}_{k't}^{\text{DC},g} y_{k't}^{\text{green}} - \sum_{f,m,c,p,v} x_{fk'mcpt}^v$ .
- *Supplier capacity* (14):  $\sigma_{iat} \perp S_{iat}^{\text{max}} - \sum_{f,j,v} x_{ifjat}^v$ .
- *Mode capacity* (17):  $\psi_{vt} \perp \text{Cap}_{vt}^{\text{tran}} - \sum_{\text{all arcs on } (v,t)} x_{vt}^v$ .
- *Award gate*:  $\pi_{iat} \perp \bar{M}_{iat} y_{iat}^{\text{sup}} - \sum_{f,j,v} x_{ifjat}^v$ .
- *Sweeping gate*:  $\varpi_{ijat} \perp \bar{M}_{ijat} w_{ijat} - \sum_{f,v} x_{ifjat}^v$ .

(Activation binaries  $y^{\text{PL}}$ ,  $y^{\text{DC}}$ ,  $y^{\text{green}}$  and awards  $y^{\text{sup}}$  are fixed in the lower level; their KKT appear in the upper-level MPEC if they are decision variables.)

#### B.5. Service proxy inside the KKT

If the service proxy  $\ell$  in the demand  $D(P, \ell)$  is endogenized (rather than treated as a parametric input from the upper level), include the linearized definitions (19)–(23) via auxiliary variables  $N_{fmct}^v$ ,  $T_{fmct}$ ,  $S_{fmct}^v$ ,  $s_{fmct}$  and linearization constraints:

- Flow aggregations (21)–(22) become equalities in  $Az = b$ .

- Shares  $S^v = N^v/T$  and  $s = \min 1, T/(C + \epsilon)$  are linearized as in the approach (SOS2 or McCormick) and enter as additional equalities/inequalities with their own duals. Their stationarity conditions follow the same reduced-cost template and need not be written out term-by-term; solvers handle them once the linearization is explicit. Linearizing  $S^v = N^v/T$  (exact MILP with McCormick+SOS2 or convex relaxation)

Introduce  $S^v \in [0,1]$ ,  $T \in [0, \bar{T}]$  and the identity  $N^v = S^v T$  (bilinear). Options:

- *Piecewise linear (SOS2) exact*: segment  $T \in [0, \bar{T}]$  at breakpoints  $0 = \tau_0 < \dots < \tau_K = \bar{T}$ . Use convex combinations  $T = \sum_k \lambda_k \tau_k$ ,  $\sum_k \lambda_k = 1$ , SOS2 on  $\lambda$ . Enforce  $N^v = \sum_k \lambda_k \tau_k S^v$ . This is linear with SOS2 and is exact on the grid.
- *McCormick envelopes (convex)*: with known bounds  $0 \leq S^v \leq 1$ ,  $0 \leq T \leq \bar{T}$ , add four linear inequalities for each bilinear product  $N^v = S^v T$ . This yields a convex outer approximation; refine with cuts or more breakpoints if needed.
- Add  $T = \sum_v N^v$  and  $\sum_v S^v \leq 1$  (the latter is implied when  $T > 0$ ; include  $S^v \leq 1$  and  $T \geq 0$ ). Linearizing  $\min 1, u$  with  $u = T/(C + \epsilon)$ , Let  $C^+ = C + \epsilon$  (constant once activations are fixed). Introduce  $0 \leq s \leq 1$  and enforce  $s \leq \frac{T}{C^+}$ ,  $s \leq 1$ .
- *Piecewise linear epigraph*: segment  $T \in [0, \bar{T}]$ , use SOS2 to enforce  $s \leq T/C^+$  on each segment. If exactness is desired, impose equality  $s = \min 1, T/C^+$  via two linear inequalities  $s \leq 1$  and  $s \leq T/C^+$  together with a convex, increasing penalty in the objective (or binary “kink” selection with big-M for strict equality). In practice,  $s = \min 1, T/C^+$  with the two inequalities is tight once  $\ell$  enters the objective or constraints. Finally,  $\ell = \alpha_0 + \alpha_1(1 - s) + \alpha_2 \sum_v \theta_v S^v$  is linear once  $S^v$  and  $s$  are linearized.
- The effective price  $P_{f_{mct}}^{\text{price}} = P_{f_{mct}} + O_{mct}(1 - \ell_{f_{mct}})$  then appears only in the RHS of market clearing  $b(P, \ell)$ , preserving the KKT envelope used in the price FOCs. Parity and smoothness constraints (43)–(45) are linear inequalities. Price bounds (33) linear.

This preserves linearity (SOS2) or convexity (McCormick) of the lower level, so the KKT system remains valid (LP/MILP case) or can be written as an MCP (convex case).

### B.6. Two equivalent treatments of carbon in KKT

Multiply each arc/production unit by its emission factor and price it at  $\tau_t$  (or the endogenous permit shadow, if pooled). These terms sit in  $c$ , so no extra dual is needed, and all stationarity equations above already include carbon. Alternatively, keep an equality for emissions  $E_{ft} - \sum(\text{factors} \times \text{flows}) = 0$  with dual  $\kappa_{ft}$  (NC/HYB) or pooled  $E_t^{\text{coal}}$  with dual  $\kappa_t$  (COOP). Then each stationarity picks up a term  $-\kappa \cdot \text{factor}$ , and  $\kappa$  is the endogenous carbon shadow price. If cap/trade constraints are also present at the lower level, add their complementary-slackness pairs (period caps, banking recursion) with the usual free/positive duals. Because demand enters only on the RHS of market clearing, Danskin’s theorem gives

$$\frac{\partial C_f^{\text{oper}}}{\partial P_{f_{mct}}} = \sum_{\rho(11)} y_{\rho} \frac{\partial D_{f_{mct}}(P, \ell)}{\partial P_{f_{mct}}} = \mu_{f_{mct}} \frac{\partial D_{f_{mct}}}{\partial P_{f_{mct}}} \quad (\text{KKT.13})$$

with the sign convention in the profit objective yielding the minus sign used in section 3.2.1–section 3.2.3. This identity is exactly what produces the NC/HYB Lerner rules and the COOP coordinated markup once we differentiate the upper-level objectives.

### Single-level reformulations (MPCC/MPEC/EPEC)

- **NC:** Replace the lower level by KKT (KKT-1)–(KKT-3); keep each firm’s price choice (or its log-price) as an upper variable and enforce the first-order condition (55)/(56) or let the solver find prices by maximizing the firm’s profit with KKT embedded. This yields an MPCC.
- **COOP:** Replace the common lower level by KKT and maximize the coalition’s objective; prices can be obtained either from (68)–(69) or as upper variables in the same MPCC. This is a single-leader MPEC.
- **HYB:** Embed each firm’s lower-level KKT and both firms’ price vectors; the Stage-1 game becomes an EPEC. Under Rosen’s condition (diagonal dominance) we have uniqueness, and you may also solve by best responses using the closed-form Lerner (73) with the KKT providing  $\mu^{\text{HYB}}$ .

*Complementarity handling.* Keep orthogonality  $a \perp b$  directly and use an MCP solver (PATH), or linearize with Fortuny-Amat–McCarl binaries  $\zeta$  and tight big- $M$  bounds:

$$0 \leq a \leq M^a \zeta, 0 \leq b \leq M^b (1 - \zeta). \quad (\text{KKT.14})$$

Choose  $M$ s from physical maxima (capacities, costs) and scale rows so  $\|g\|_\infty \leq 1$ .

## Appendix CC. Detailed Case Study Data and Parameter Calibration

The appendix reports detailed firm-specific information on production facilities, distribution centers, supplier pools, transportation costs and emissions, carbon policy parameters, and investment options for the two competing firms operating in Türkiye. Tables C.1–C.3 summarize, respectively, the firm-specific parameters for Firm 1, Firm 2, and the system-wide market, transportation, and policy parameters shared by both firms.

| Category    | Parameter                                 | Symbol                                      | Firm 1 value                                                          | Unit                     | Notes                                                        |
|-------------|-------------------------------------------|---------------------------------------------|-----------------------------------------------------------------------|--------------------------|--------------------------------------------------------------|
| Production  | Unit production cost (A/B)                | $(c_{1jp}^{\text{prod}})$                   | P1 Istanbul: 42 / 48; P2 Ankara: 44 / 50; P3 Gaziantep: 43 / 49       | TRY/unit                 | Case data                                                    |
|             | Nominal production capacity (A/B)         | $(\overline{Q}_{1jp}^{\text{prod}})$        | P1: 28k / 22k; P2: 24k / 20k; P3: 20k / 18k                           | units/period             | Case data                                                    |
|             | Expandable capacity (A/B)                 | $(\Delta \overline{Q}_{1jp}^{\text{prod}})$ | P1: +8k / +6k; P2: +7k / +5k; P3: +6k / +5k                           | units/period             | Discrete option                                              |
|             | Capacity expansion CAPEX                  | $(K_{1jp}^{\text{cap}})$                    | P1: 3.2; P2: 2.8; P3: 2.2                                             | M TRY                    | Case data                                                    |
|             | Green retrofit emission reduction         | $(\rho_{1jp}^{\text{prod}})$                | 0.28; 0.25; 0.27                                                      | –                        | Case data                                                    |
|             | Green retrofit investment                 | $(K_{1jp}^{\text{green}})$                  | 4.1; 3.6; 3.1                                                         | M TRY                    | Case data                                                    |
|             | Production emission factor (pre-retrofit) | $(e_{1jp}^{\text{prod}})$                   | Calibrated: A = 1.00, B = 1.08                                        | kg CO <sub>2</sub> /unit | Typical light manufacturing in TR; reduced by $((1 - \rho))$ |
| Private DCs | DC handling capacity (A/B)                | $(\overline{Q}_{1kp}^{\text{DC}})$          | D1 Istanbul: 30k / 24k; D2 Adana: 24k / 20k; D3 Diyarbakır: 18k / 15k | units/period             | Case data                                                    |
|             | Inventory holding cost                    | $(c_{1kp}^{\text{inv}})$                    | 1.25; 1.10; 1.05                                                      | TRY/unit·period          | Case data                                                    |

|                  |                                             |                              |                        |                          |                              |
|------------------|---------------------------------------------|------------------------------|------------------------|--------------------------|------------------------------|
| Shared green DC  | Pooled handling capacity                    | $(\overline{Q}_{k'}^{DC,g})$ | G1 Konya: 40,000       | units/period             | Case data                    |
|                  | Holding/handling cost                       | $(c_{1k'p}^{Inv})$           | 1.40                   | TRY/unit·period          | Case data                    |
|                  | Transport emission reduction                | $(\rho_k^{Trans})$           | 0.35                   | –                        | Adjacent legs $\times$ 0.65  |
|                  | Fixed activation cost                       | $(F_{k'}^g)$                 | 6.0                    | M TRY                    | Shared via $(\lambda_{fk'})$ |
| Suppliers        |                                             |                              |                        |                          |                              |
|                  | Domestic purchase price (A/B)               | $(c_{iat}^{wh})$             | 17–19 / 20–23          | TRY/unit                 | Ex-works                     |
|                  | Import purchase price (A/B)                 | $(c_{iat}^{wh})$             | 15–16 / 18–19          | EUR/unit                 | FX & tariffs applied         |
|                  | Import tariff                               | $(\tau_i)$                   | 0.03–0.07              | fraction                 | Case data                    |
|                  | Supplier capacity                           | $(S_{iat}^{max})$            | 16k–26k                | units/period             | Case data                    |
|                  | Lead time                                   | $(L_{ait})$                  | 2–3 (dom.), 5–7 (int.) | days                     | Case data                    |
|                  | Quality score                               | $(q_{ai})$                   | 0.78–0.86              | –                        | Case data                    |
|                  | Supplier emission factor                    | $(e_{ia}^{sup})$             | 0.20–0.45              | kg CO <sub>2</sub> /unit | Case data                    |
|                  | Green supplier reduction                    | $(\rho_{ia}^{sup})$          | 0.15                   | –                        | Case data                    |
| Policy & finance | Domestic sourcing floor (primary/secondary) | $(\rho_{1a}^{dom})$          | 0.55 / 0.40            | –                        | Case data                    |
|                  | Plant×product activation cost               | $(F_{1jp}^{PL})$             | 0.9–1.4                | M TRY/period             | Case data                    |
|                  | DC activation cost                          | $(F_{1k}^{DC})$              | 0.8–1.2                | M TRY/period             | Case data                    |
|                  | Investment budget                           | $(B_{1t})$                   | 18                     | M TRY/period             | Case data                    |
|                  | Minimum price (margin floor)                | $(M_{1mct})$                 | Cost + 8–10 %          | TRY/unit                 | Policy constraint            |

| Table C2. Firm 2 (Türkiye) – Model parameters and calibrated data |                                   |                                      |                                      |              |                 |
|-------------------------------------------------------------------|-----------------------------------|--------------------------------------|--------------------------------------|--------------|-----------------|
| Category                                                          | Parameter                         | Symbol                               | Firm 2 value                         | Unit         | Notes           |
| Production                                                        | Unit production cost (A/B)        | $(c_{1jp}^{prod})$                   | P4 Izmir: 43 / 47; P5 Bursa: 45 / 51 | TRY/unit     | Case data       |
|                                                                   | Nominal production capacity (A/B) | $(\overline{Q}_{1jp}^{prod})$        | P4: 25k / 21k; P5: 22k / 19k         | units/period | Case data       |
|                                                                   | Expandable capacity (A/B)         | $(\Delta \overline{Q}_{1jp}^{prod})$ | P4: +7k / +5k; P5: +6k / +5k         | units/period | Discrete option |
|                                                                   | Capacity expansion CAPEX          | $(K_{1jp}^{cap})$                    | 3.0; 2.5                             | M TRY        | Case data       |

|                  |                                             |                              |                                           |                          |                                         |
|------------------|---------------------------------------------|------------------------------|-------------------------------------------|--------------------------|-----------------------------------------|
|                  | Green retrofit emission reduction           | $(\rho_{1jp}^{prod})$        | 0.26; 0.24                                | –                        | Case data                               |
|                  | Green retrofit investment                   | $(K_{1jp}^{green})$          | 3.9; 3.3                                  | M TRY                    | Case data                               |
|                  | Production emission factor (pre-retrofit)   | $(e_{1jp}^{prod})$           | Calibrated: A = 0.98, B = 1.06            | kg CO <sub>2</sub> /unit | Slightly lower due to coastal logistics |
| Private DCs      | DC handling capacity (A/B)                  | $(\overline{Q}_{1kp}^{DC})$  | D4 Izmir: 26k / 22k; D5 Ankara: 20k / 18k | units/period             | Case data                               |
|                  | Inventory holding cost                      | $(c_{1kp}^{Inv})$            | 1.20; 1.15                                | TRY/unit·period          | Case data                               |
| Shared green DC  | Pooled capacity                             | $(\overline{Q}_{k'}^{DC,g})$ | G1 Konya: 40,000                          | units/period             | Shared                                  |
|                  | Holding/handling cost                       | $(c_{1k'p}^{Inv})$           | 1.40                                      | TRY/unit·period          | Case data                               |
|                  | Transport emission reduction                | $(\rho_{k'}^{Trans})$        | 0.35                                      | –                        | Adjacent legs × 0.65                    |
|                  | Cost-sharing weight                         | $(F_{k'}^g)$                 | Baseline: 0.50                            | –                        | Equal share benchmark                   |
| Suppliers        |                                             |                              |                                           |                          |                                         |
|                  | Domestic purchase price (A/B)               | $(c_{iat}^{wh})$             | 18–19 / 21–23                             | TRY/unit                 | Case data                               |
|                  | Import purchase price (A/B)                 | $(c_{iat}^{wh})$             | 15–17 / 18–20                             | EUR/unit                 | Case data                               |
|                  | Import tariff                               | $(\tau_i)$                   | 0.03–0.07                                 | fraction                 | Case data                               |
|                  | Supplier capacity                           | $(S_{iat}^{max})$            | 18k–22k                                   | units/period             | Case data                               |
|                  | Lead time                                   | $(L_{ait})$                  | 2–3 (dom.), 6–7 (int.)                    | days                     | Case data                               |
|                  | Quality score                               | $(q_{ai})$                   | 0.79–0.85                                 | –                        | Case data                               |
| Policy & finance | Domestic sourcing floor (primary/secondary) | $(e_{ia}^{sup})$             | 0.50 / 0.40                               | –                        | Case data                               |
|                  | Plant×product activation cost               | $(\rho_{ia}^{sup})$          | 0.9–1.4                                   | M TRY/period             | Case data                               |
|                  | DC activation cost                          | $(\rho_{1a}^{dom})$          | 0.8–1.2                                   | M TRY/period             | Case data                               |
|                  | Investment budget                           | $(F_{1jp}^{PL})$             | 16                                        | M TRY/period             | Case data                               |
|                  | Minimum price (margin floor)                | $(F_{1k}^{DC})$              | Cost + 8–10 %                             | TRY/unit                 | Policy constraint                       |

**Table C3.** System-wide and market parameters (common to both firms)

| Category | Parameter | Symbol | Value | Unit | Notes (Türkiye-realistic) |
|----------|-----------|--------|-------|------|---------------------------|
|----------|-----------|--------|-------|------|---------------------------|

|                     |                                |                                          |                                                                                 |                              |                              |
|---------------------|--------------------------------|------------------------------------------|---------------------------------------------------------------------------------|------------------------------|------------------------------|
| Demand              | Base demand scale              | $(\theta_{mct})$                         | TR-UR: 14k/9k; TR-UE: 12.5k/8.5k; TR-RR: 8k/5k; DE-EC: 9.5k/6k; ME-R: 6.5k/4.5k | units/period                 | Case data                    |
|                     | Own-price elasticity           | $(\beta_{mct})$                          | 1.35–1.85 (by market/channel)                                                   | –                            | Diagonal dominance preserved |
|                     | Cross-price elasticity         | $(\eta_{mct}^{ff'})$                     | 0.20–0.30                                                                       | –                            | Symmetric                    |
|                     | Brand/service intercept        | $(\alpha_0)$                             | 0.82                                                                            | –                            | Case data                    |
|                     | Service sensitivity            | $(\alpha_1, \alpha_2)$                   | 0.30; 0.08                                                                      | –                            | Case data                    |
|                     | Mode penalty                   | $(\theta_v)$                             | Truck 0.08; Rail 0.04; Ship 0.06                                                | –                            | Reliability effect           |
|                     | Service penalty weight         | $(O_{mct})$                              | TR 4.0; DE 5.5; ME 4.5                                                          | TRY                          | Case data                    |
| Transport costs     | Unit transport cost            | $(c_{vuh}^{Tran})$                       | Truck 2–7; Rail 2–5; Ship 1.8–3.8                                               | TRY/unit                     | Türkiye/EU corridors         |
| Transport emissions | Mode emission factor           | $(e_{vuh}^{Tran})$                       | Truck 0.26; Rail 0.08–0.10; Ship 0.05–0.07                                      | kg CO <sub>2</sub> /unit·leg | IPCC-consistent              |
| Transport capacity  | Mode capacity                  | $(Cap_{vt}^{tran})$                      | Truck 60k; Rail 45k; Ship 35k                                                   | units/period                 | Case data                    |
| Inventory           | Spoilage rate                  | $(\phi)$                                 | 0.1 %                                                                           | %/period                     | Negligible                   |
| Carbon policy       | Carbon tax (baseline)          | $(\tau_t)$                               | 1,200                                                                           | TRY/tCO <sub>2</sub>         | Case value                   |
|                     | Carbon permit price (baseline) | $(\pi_t)$                                | 1,300                                                                           | TRY/tCO <sub>2</sub>         | Case value                   |
|                     | Emission quota                 | $(\widehat{E_{ft}})$                     | Firm 1: 120; Firm 2: 110                                                        | tCO <sub>2</sub> /period     | Case data                    |
|                     | Trading bounds                 | $(v_{ft}^{min}, v_{ft}^{max})$           | –40, +60                                                                        | tCO <sub>2</sub> /period     | Case data                    |
| Pricing rules       | Price bounds                   | $(\underline{p}_{fmct}, \bar{p}_{fmct})$ | As specified per market/channel                                                 | TRY/unit                     | Case data                    |
|                     | Max price change               | $(\Delta P_{mc}^{max})$                  | 7–10                                                                            | TRY/period                   | Case data                    |
|                     | Channel price gap              | $(\Delta f_{mtch})$                      | 6 (dom.), 8 (export)                                                            | TRY                          | Case data                    |

## 5.2. Computational Certification, Robustness, and $\epsilon$ -Equilibrium Verification

We now summarize the computational certification properties of the proposed solution approach and assess its robustness with respect to initialization, trust-region design, and numerical equilibrium accuracy. This subsection complements the case study results by formally documenting optimality guarantees, computational stability, and equilibrium verification checks.

### *Optimality gaps and computational performance*

Across all scenarios NC, HYB, and COOP the proposed framework delivers solutions with certified near-optimality. As reported in Tables D1–D3, the strategic master problem terminates with optimality gaps consistently below 0.25%, while the baseline monolithic formulation exhibits substantially larger gaps ( $\approx 1\%$ ). Wall-clock times are decomposed into upper-level (strategic master) and lower-level (operational oracle) components, reflecting the bilevel structure of the problem. Median wall-times across multiple seeds remain within practical limits for all regimes, demonstrating the scalability of the proposed decomposition with equilibrium oracles. Importantly, the tighter gaps achieved by the proposed approach coincide with *lower* computational times relative to the baseline, confirming that improved solution quality is not obtained at the expense of computational burden.

### *Robustness to initialization*

Robustness with respect to initialization is assessed by solving each scenario under multiple random seeds and reporting median outcomes and dispersion measures (stability columns in Tables D1–D3). Across all regimes, dispersion in profit, residuals, and solve times remains below 2%, and the economic ranking of governance regimes ( $\text{HYB} \geq \text{COOP} \geq \text{NC}$  in profits,  $\text{COOP} \leq \text{HYB} \leq \text{NC}$  in emissions) is invariant. This confirms that results are not driven by favorable starting points or path-dependent artifacts of the solution process.

### *Trust-region design and algorithmic sensitivity*

The strategic master problem employs an adaptive binary trust-region, with a dynamically adjusted Hamming-ball radius bounded within [2, 12]. This mechanism stabilizes exploration of discrete design decisions without restricting convergence to high-quality solutions. Algorithmic sensitivity is examined through an ablation study (Table D15), comparing performance with and without DCEC. While the inclusion of DCEC substantially reduces the number of master iterations and wall-time and tightens final optimality gaps, solution values and regime rankings remain unchanged. This demonstrates that the reported results are robust to trust-region settings and cut management strategies and do not rely on fine-tuned parameter choices.

### *$\epsilon$ -equilibrium definition and verification*

For the NC and HYB regimes, strategic pricing is determined through mixed complementarity formulations. Solutions are therefore delivered as  $\epsilon$ -equilibria, where  $\epsilon$  is defined as the maximum unilateral profit deviation across all firms and market–channel tuples, computed via the MCP natural residual.  $\epsilon = \max \{ \| F(z) \|_\infty \}$ , with  $F(z)$  denoting the MCP residual mapping. Across all reported instances,  $\epsilon$  remains below  $10^{-6}$ , as documented in Tables D1–D3. For the COOP regime, equilibrium conditions are verified through KKT residuals of the centralized pricing problem, which are likewise below  $10^{-6}$  at termination.

Additional verification checks include: (i) primal and dual feasibility of all lower-level LPs within  $10^{-8}$  tolerances, (ii) consistency between realized flows and market-clearing conditions, and (iii) stability of objective values and multipliers upon re-solving with alternative solvers (IPOPT vs. PATH for pricing oracles, CPLEX vs. Gurobi for LPs; see Table D15). Together, these checks ensure that no firm can unilaterally deviate to improve its payoff beyond numerical tolerance, and that reported outcomes represent economically meaningful equilibria rather than numerical artifacts.

| <b>Table D15. Algorithmic ablation and Benchmark Comparison</b> |              |             |                        |               |
|-----------------------------------------------------------------|--------------|-------------|------------------------|---------------|
| Configuration                                                   | Master iters | Time (min)  | Final gap (%)          | UB difference |
| Standard LBB (single optimality cuts)                           | 28           | 9.6         | 0.92                   | −1.14 M TRY   |
| LBB without equilibrium oracles                                 | 24           | 8.1         | 0.71                   | −0.78 M TRY   |
| LBB-EO (no DCEC)                                                | 19           | 6.8         | 0.48                   | −0.23 M TRY   |
| LBB-EO + DCEC (proposed)                                        | 10           | 4.1         | 0.21                   | baseline      |
| COOP price oracle: IPOPT                                        | —            | 2.2         | KKT $7 \times 10^{-7}$ | same UB       |
| COOP price oracle: PATH-NLP                                     | —            | 2.9         | KKT $8 \times 10^{-7}$ | same UB       |
| Follower LP: CPLEX vs Gurobi                                    | —            | 1.0 vs 1.05 | —                      | same UB       |

#### 5.4. Sensitivity Analysis

Table D16 in Appendix D extends robustness to the calibration levers explicitly service proxy parameters ( $\alpha_2, \theta_v$ ) and procurement scoring weights ( $\lambda_\ell, \lambda_e$ ), and confirms that HYB/COOP remain more responsive than NC because they can exploit pooling, mode shifting, and the shared green corridor, yielding larger profit gains and larger emission reductions for the same parameter shift. Finally, Table D17 in Appendix D provides ablation checks that rule out a double-counting interpretation  $\lambda_e$  acts as a procurement screening preference, whereas  $\tau$  cap-and-trade price realized emissions; switching each channel on/off produces distinct and interpretable changes in profit and emissions, while preserving qualitative scenario insights.

The results reported in Table D16 confirm that the main economic–environmental trade-offs of the model are robust to simultaneous perturbations of the parameters ( $\alpha, \theta, \lambda$ ). Increasing the service–price coupling parameters  $\alpha_2$  and  $\theta_v$  leads to systematic profit improvements under HYB and COOP, while NC exhibits only marginal responses. Quantitatively, HYB shows the largest profit sensitivity, reflecting its ability to monetize service improvements through independent pricing, whereas COOP translates the same changes primarily into additional abatement rather than profit. Variations in the carbon- and lead-time–related weights ( $\lambda_e, \lambda_\ell$ ) generate limited profit volatility (sub-1% across all regimes) but induce material emission reductions, especially under COOP, where centralized routing and pricing compress the shadow cost of the emission constraint. HYB remains close to COOP in emission response but preserves a small profit advantage, consistent with the baseline findings. Importantly, no parameter realization reverses the regime ranking  $\text{HYB} \geq \text{COOP} \geq \text{NC}$  in profit, nor does any perturbation induce unstable oscillations or binding conflicts between carbon tax and quota mechanisms. These results demonstrate that ( $\alpha, \theta, \lambda$ ) primarily scale the strength of observed mechanisms service monetization, pooled routing, and permit banking without altering their direction, confirming the structural robustness of the proposed formulation to calibration uncertainty in critical parameters.

Table D17 shows when  $\lambda_e$  is switched off, profits rise marginally but emissions deteriorate, showing that supplier screening contributes incremental abatement beyond the priced-carbon mechanism. Conversely, removing priced carbon ( $\tau$  and cap/trading) increases profits but increases emissions substantially, indicating that compliance instruments are the primary driver of monetized abatement. Across both ablations, the qualitative scenario ranking and the operational mechanisms (mode shifting and hub consolidation) remain stable, supporting robustness of the main conclusions.

| <b>Table D16.</b> Parametric Sensitivity of Profit and Emissions to Service, Demand, and Carbon Cost Couplings |                                                                                 |                                               |                                                |                                                 |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| Targeted check                                                                                                 | Meaning                                                                         | NC: $\Delta\text{Profit} / \Delta\text{Emis}$ | HYB: $\Delta\text{Profit} / \Delta\text{Emis}$ | COOP: $\Delta\text{Profit} / \Delta\text{Emis}$ |
| $(\alpha_2 + 15\%)$                                                                                            | stronger mode-quality impact on service                                         | +0.10 / -0.20                                 | +0.35 / -0.60                                  | +0.30 / -0.70                                   |
| $(\theta_{rail} + 20\%)$                                                                                       | rail seen as more reliable $\rightarrow$ service improves when shifting to rail | +0.05 / -0.10                                 | +0.25 / -0.45                                  | +0.22 / -0.55                                   |
| $(\theta_{ship} + 20\%)$                                                                                       | ship reliability weight up on export lanes                                      | +0.04 / -0.08                                 | +0.18 / -0.35                                  | +0.16 / -0.40                                   |
| Re-weight score toward lead time: $(\lambda_1 \uparrow)(renorm)$                                               | awards favor faster suppliers                                                   | +0.03 / -0.10                                 | +0.10 / -0.25                                  | +0.08 / -0.22                                   |
| Re-weight score toward supplier carbon: $(\lambda_e \uparrow)(renorm)$                                         | awards favor lower $(e_{ia}^{sup})$ suppliers                                   | -0.02 / -0.18                                 | -0.05 / -0.45                                  | -0.06 / -0.55                                   |

| <b>Table D17.</b> System Performance Under Combined Market, Capacity, and Carbon Policy Shocks |                                                       |                                               |                                                |                                                 |
|------------------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| Ablation                                                                                       | What it isolates                                      | NC: $\Delta\text{Profit} / \Delta\text{Emis}$ | HYB: $\Delta\text{Profit} / \Delta\text{Emis}$ | COOP: $\Delta\text{Profit} / \Delta\text{Emis}$ |
| $Set(\lambda_e = 0)(keep(\tau, \hat{E}, \pi)active)$                                           | removes <i>screening</i> channel, keeps priced carbon | +0.03 / +0.10                                 | +0.06 / +0.25                                  | +0.05 / +0.30                                   |
| $Set(\tau = 0)andremovecap / trading(keep(\lambda_e > 0))$                                     | removes <i>priced carbon</i> , keeps screening        | +0.40 / +1.10                                 | +0.55 / +1.40                                  | +0.50 / +1.60                                   |

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